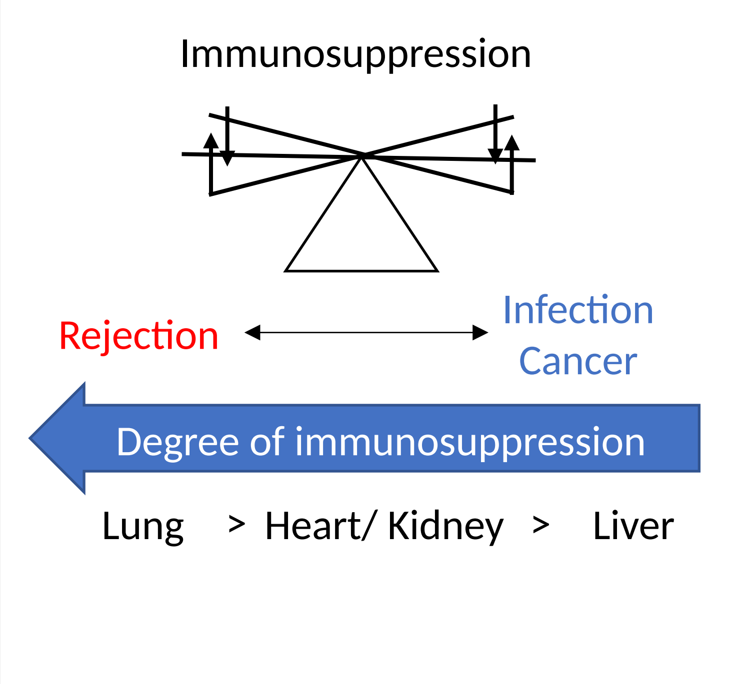
TRANSPLANT MEDICINE

Oversuppression

Undersuppression

Click anywhere to continue.



Balancing immunosuppression

Induction vs. maintenance

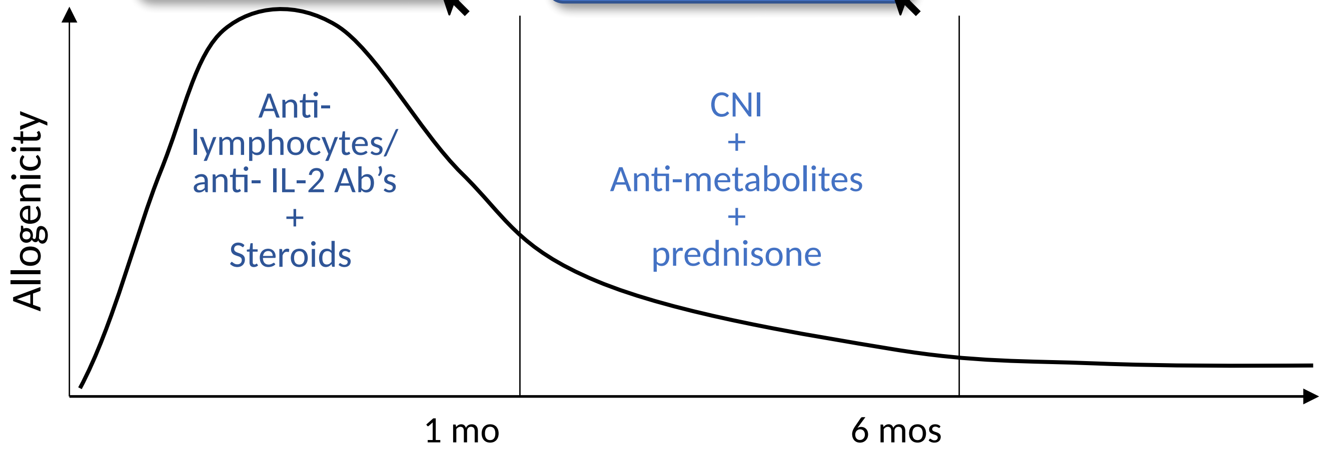
Complications

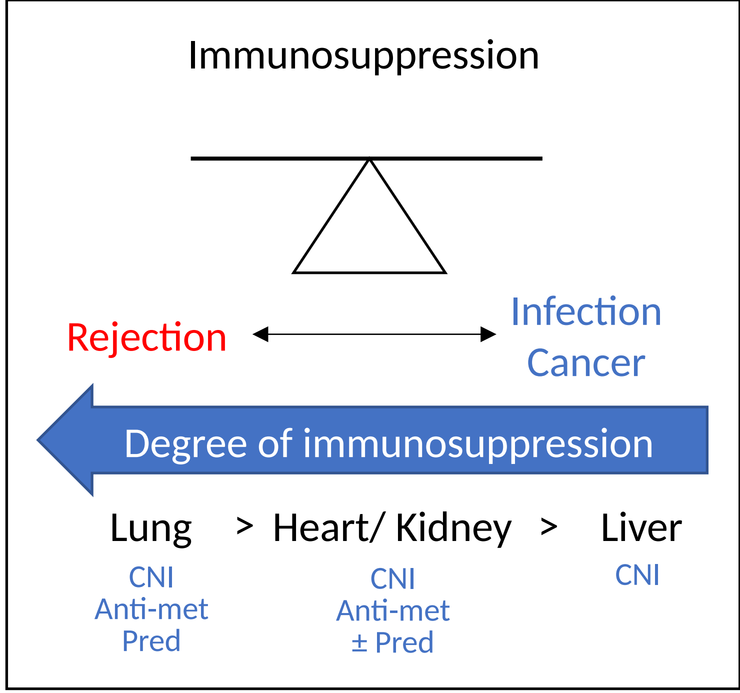
TRANSPLANT MEDICINE

INDUCTION

MAINTENANCE

MEDICATIONS





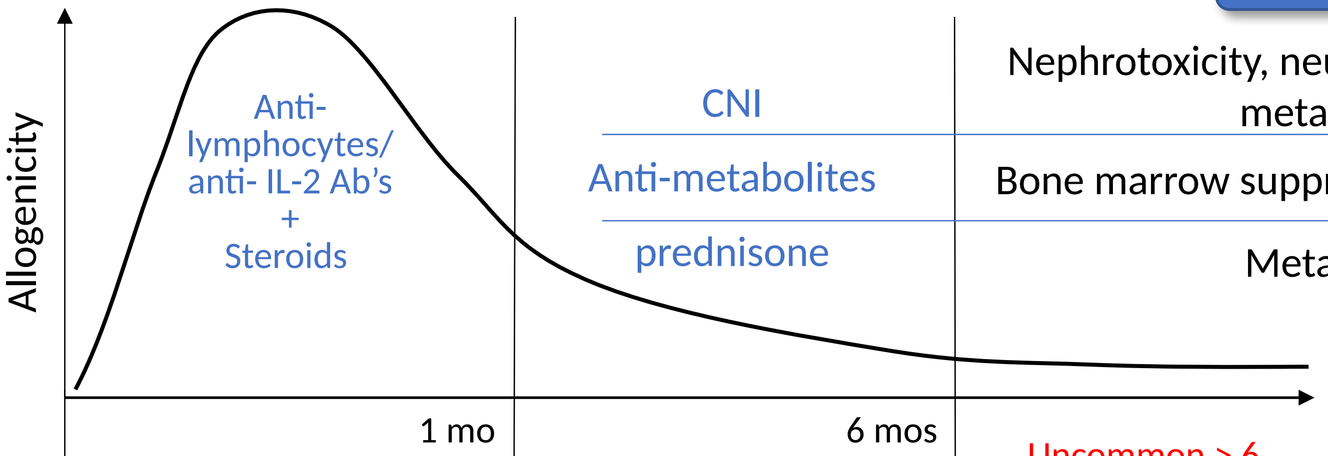
Click anywhere to continue.

Balancing immunosuppression

Induction vs. maintenance

Complications

TRANSPLANT MEDICINE



	INDUCTION	MAINTENANCE	Medication Effects
MEDICATIONS	Anti- lymphocytes/ anti- IL-2 Ab's + Steroids	CNI	Nephrotoxicity, neurotoxicity, ↑ K and uric acid, metabolic syndrome
		Anti-metabolites	Bone marrow suppression, GI side effects (colitis)
		prednisone	Metabolic syndrome

COMPLICATIONS

	0–1 mo	1–6 mos	> 6 mos
Rejection	~ 25% with acute rejection		Uncommon > 6 mos unless med non-compliance
		Chronic rejection	Chronic rejection
Infections	Donor/ recipient + Nosocomial	Reactivation (TB, HBV, HSV/VZV, CMV, BK) + OIs	Community acquired >> OIs (molds, Mucor, late viral)
Malignancy	Overall 2x as likely.	PTLD (multiorgan/ SB > lung > kidney/heart > liver)	Skin cancer, anogenital
Surgical	Leaks and kinks	Fistulas and stenoses	

Newer Context (2011–2023)

Belatacept (Nulojix, FDA 2011) — CNI-sparing maintenance

CD28/CD80-86 costimulation blocker; better long-term GFR & fewer donor-specific Abs vs CNI (BENEFIT). CONTRAINDICATED if EBV-seronegative (PTLD/CNS-PTLD, PML risk).

Induction depleting agents have shifted

OKT3 / muromonab-CD3 discontinued 2010; current depleting agents = rabbit ATG, alemtuzumab. Non-depleting = basiliximab (anti–IL-2R).

CMV prophylaxis options expanded

Letermovir (Prevymis) FDA-approved Jun 2023 for high-risk (D+/R–) kidney transplant — alternative to valganciclovir with less myelosuppression. Maribavir (Livtencity, FDA 2021) for refractory/resistant CMV.

Sources: Archdeacon Am J Transplant 2012; FDA/Merck Jun 2023; LiverTox/Muromonab-CD3 2010. Standing framework (induction vs maintenance, complication timeline) unchanged.

Take-Home Points

Immunosuppression is a constant balance: less rejection vs. more infection, cancer, and drug toxicity.

Allogenicity (rejection risk) is highest in the first month and falls over time — so does the intensity of immunosuppression. Lung > heart/kidney > liver.

Assuming adherence, a FIRST episode of acute rejection beyond 6 months is highly unlikely.

Reactivation infections (CMV, BK, HSV/VZV, TB, HBV) cluster at 1–6 months; nosocomial/donor-derived <1 month; community-acquired >6 months.

PTLD occurs early (first year, EBV-driven); most other malignancies appear late. Overall cancer risk ~2× general population.

Attribution & Changelog

Adapted from TeachIM.org — “Transplant Medicine” (Yilin Zhang MD, Brandon Fainstad MD; published Dec 2020).

Optimized 2026-06-11. Corrections (learner board): “mycophenylate”→“mycophenolate”; “bortezumab”→“bortezomib”. Presenter board: no text errors found.

Added a newer-context slide (belatacept; OKT3 discontinued; letermovir/maribavir CMV) — standing teaching unchanged. All original click-to-reveal animations/graphics preserved.

Companion files: Transplant-Medicine-Handout.docx; Transplant-Medicine-Learner-Board-1.pptx (corrected). Educational use only — clinician-review before teaching.